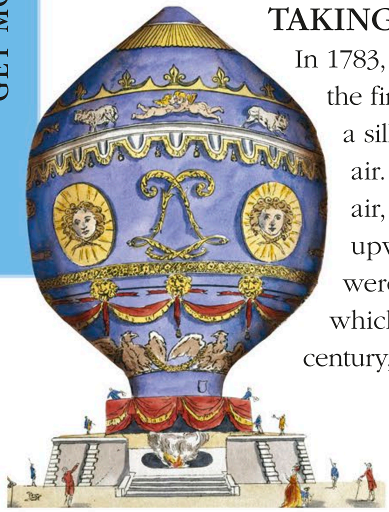
Other flying machines

Planes rely on wings to fly, but they must fly forward nonstop for the wings to produce “lift.” Other ways of flying include the use of whirling rotor blades that generate lift in helicopters simply by spinning around. That’s why helicopters and drones can take off and land almost vertically, and hover in midair.



TAKING TO THE SKIES

In 1783, the Montgolfier brothers made the first successful human flight, in a silk balloon (left) filled with hot air. Hot air is lighter than cool air, and it made the balloon rise upward. Other early balloons were filled with hydrogen gas, which is lighter than air. For a century, people flew using balloons.

Replica of the Bell 47 helicopter

SPINNING BLADES

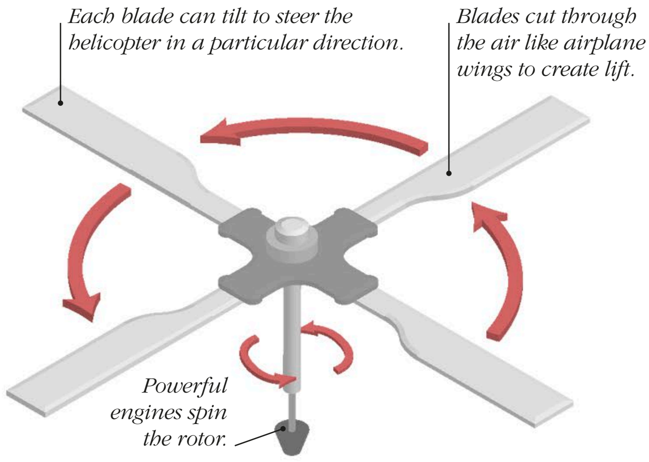
Used mainly by the armed forces in the beginning, helicopters were developed in the 1920s by the German engineer Anton Flettner and the Russian airplane designer Igor Sikorsky, among others. Helicopters really took off in 1946 with the Bell 47, made for civilian use. It had a cleverly balanced two-blade rotor, which made it compact and stable.

GIANTS IN THE AIR

A hundred years ago, large airships carried passengers in luxury across the Atlantic Ocean. Like balloons, airships are lifted by a lighter-than-air gas, such as helium, but they also have engines to help them fly in any direction. Today, the largest aircraft in the world is the Airlander 10 airship (below), with a length of 302 ft (92.05 m). This experimental craft doesn’t need a runway and can carry heavy cargo to remote places.

FORCES IN FLIGHT

As a helicopter’s rotor spins, it creates lift. With a collective pitch control, the pilot can increase the angle or “pitch” of all the blades at once to get more lift.



Helicopter rotor blades